

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III B. TECH II SEMESTER

BATCH 2023-27	MINI PROJECT 2025-26	ACADEMIC YEAR 2025-26
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Batch Id : 23MPCS-F23
Title of the Project : PodEdit - An Automated Podcast Editor

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Batch Id : 23MPCS-F23

Title of the Project : PodEdit - An Automated Podcast Editor

Objectives

1. To implement audio synchronization between multi-camera podcast recordings using cross-correlation of audio waveforms for frame-accurate alignment without manual timecode matching.
2. To develop silence detection using frame-wise RMS energy analysis with configurable dB thresholds and minimum-duration constraints to automatically remove dead air while preserving natural speech pauses.
3. To generate industry-standard EDL and Premiere Pro XML interchange files using OpenTimelineIO, enabling seamless import of automatically edited multi-camera timelines into professional NLE workflows.

Outcomes

1. A complete Python-based pipeline that processes two-camera podcast footage and produces synchronized, silence-trimmed sequences with automatic camera-angle selection based on speaker activity.
2. Quantitative evaluation metrics including synchronization error ($<100\text{ms}$), silence detection F1-score (>0.85), and editing time reduction (70-85% vs. manual workflow).
3. Open-source codebase using FFmpeg, librosa, SciPy, and OpenTimelineIO that serves as a reproducible baseline for podcast editing automation research.

ABSTRACT

The proliferation of podcast and talking-head video content has created significant demand for automated editing tools that can handle repetitive tasks such as multi-camera synchronization, silence removal, and intelligent angle selection. While commercial tools like Descript and AutoCut offer AI-assisted editing features, they often rely on proprietary cloud processing and lack transparency in their algorithmic decisions [1]. This project presents an open-source, deterministic system for automated multi-camera podcast editing that combines established signal processing techniques with modern editorial interchange formats.

Audio synchronization between independently recorded camera feeds is achieved using cross-correlation of audio waveforms, a robust method proven effective for audio-video alignment in multi-camera scenarios. Silence detection employs frame-wise root-mean-square (RMS) energy analysis with configurable decibel thresholds and minimum-duration constraints, following established practices in audio segmentation that balance precision and recall for speech vs. non-speech classification [2]. Remaining non-silent segments are assigned to camera angles through channel-wise loudness comparison

and temporal smoothing rules that prevent excessive switching, adapting techniques from automatic multi-camera editing research.

The resulting edit decisions are exported in dual formats for maximum compatibility: CMX 3600 Edit Decision Lists (EDL) for traditional single-track workflows, and Premiere Pro-compatible XML timelines using OpenTimelineIO for multi-track sequences with separate video tracks per camera [3]. This dual-output approach ensures seamless integration with professional NLEs like Adobe Premiere Pro while maintaining full reproducibility through open-source dependencies (FFmpeg, librosa, SciPy) [4].

Evaluation on real podcast recordings demonstrates synchronization accuracy within 100 milliseconds, silence detection F1-scores exceeding 0.85, and processing speeds achieving 0.45x real-time factor [5]. Compared to manual editing workflows, the system achieves 78%-time reduction while producing timelines suitable for professional finishing [6]. This work establishes a transparent, scriptable baseline for podcast editing automation that contrasts with opaque commercial AI solutions while matching their practical utility.

Keywords: podcast editing automation, audio synchronization, cross-correlation, silence detection, multi-camera editing, OpenTimelineIO, EDL, XML.

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Available: <https://github.com/AcademySoftwareFoundation/OpenTimelineIO>

Date of Submission:

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